

Web Development Concepts -- Guided Notes ANSWER KEY

Unit tp-3.10 | CompTIA Network+ N10-009 Obj. FC0-U71 3.10 | N10-008 FC0-U71 3.10

For instructor use / distribute after students complete the notes.

Question 1

HTML stands for _____ and defines the _____ (content and organization) of a web page. CSS stands for _____ and defines the _____ (colors, fonts, layout). JavaScript adds _____ to the page, making it respond to user actions without reloading.

Answer: HyperText Markup Language; structure; Cascading Style Sheets; appearance (style); interactivity (behavior).

Question 2

HTTP (HyperText Transfer Protocol) is the protocol used to request and serve _____ over the web. HTTPS is the secure version that _____ traffic using TLS. HTTP uses port _____ and HTTPS uses port _____. Browsers display a _____ icon in the address bar for HTTPS sites.

Answer: Web pages; encrypts; 80; 443; padlock.

Question 3

DNS (Domain Name System) translates human-readable _____ (like www.google.com) into _____ addresses that computers use to locate servers. Without DNS, users would need to type _____ addresses directly to reach websites. A DNS _____ attack replaces a legitimate IP with a fraudulent one to redirect users.

Answer: Domain names; IP; IP; poisoning (spoofing).

Question 4

_____ side processing runs on the web server (PHP, Python, Node.js) and sends the finished HTML page to the client. _____ side processing runs in the user's browser (JavaScript) and modifies the page without contacting the server. Modern web apps use both: the server delivers _____ and the client updates the _____ dynamically.

Answer: Server; client; initial HTML; DOM (page content).

Question 5

A _____ is a pre-built set of code libraries and tools that provides a structure for building web applications. Front-end frameworks include _____ and _____. Back-end frameworks include Django (Python) and _____. Using a framework reduces _____ by providing solved solutions for common tasks.

Answer: Framework; React, Angular (examples); Express (Node.js) or Flask; development time (redundancy).

Question 6

A web browser renders HTML, CSS, and JavaScript into the visual page a user sees. The _____ (Document Object Model) is a tree structure representing the page, which JavaScript can modify. Browsers use a _____ engine to interpret JavaScript (Chrome uses V8, Firefox uses _____). Developer _____ in the browser allow inspecting HTML, CSS, and network requests.

Answer: DOM; JavaScript; SpiderMonkey; tools (DevTools).

Question 7

A URL (Uniform Resource Locator) has several parts. In the URL

https://shop.example.com/products?id=42:

https is the _____

shop.example.com is the _____

/products is the _____

?id=42 is the _____

Answer: Protocol (scheme); domain (hostname); path; query string (parameters).

Question 8 -- Real World Application

REAL WORLD: A student builds a simple website and deploys it using only HTTP (not HTTPS). The site includes a login form where users submit their username and password. A cybersecurity teacher tells them this is dangerous. Explain why HTTP is dangerous for a login form compared to HTTPS, and describe what would happen if an attacker intercepted the traffic.

Answer: With HTTP, all data transmitted between the browser and web server is sent in plain text -- it is not encrypted. If an attacker is on the same network (e.g., cafeteria Wi-Fi) and runs a packet capture tool like Wireshark, they can see every byte of the HTTP traffic. When a user submits the login form, the username and password are transmitted as readable text in the HTTP POST request. The attacker reads the credentials directly from the captured packet. With HTTPS (TLS), the connection is encrypted with a certificate. Even if the attacker captures the packets, all they see is encrypted ciphertext -- they cannot read the credentials without the private key. The fix is to install a TLS certificate (free options: Let's Encrypt) and configure the server to redirect all HTTP requests to HTTPS.